

A Physics-Informed Neural Network Atlas: Sub-Percent on Smooth Forward PDEs, Orders Worse on Inverse, High-Frequency and Real Data

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CAOS open-research programme · PINN-Lab, physics-informed neural network method catalogue

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_PINNLAB

Abstract. Physics-informed neural networks (PINNs) are promoted as a general differential-equation solver, but the accuracy actually achieved varies by orders of magnitude across problem types, and that variation is rarely laid out in one place. This report is a method atlas: a runnable catalogue in which each of the state-of-the-art PINN families (hard constraints, residual-based adaptive sampling, Fourier features, domain decomposition, operator learning, physics-informed operators, conformal-prediction uncertainty, structure-preserving Hamiltonian learning, and inverse solving) is exercised on at least one differential-equation case, trained by a dedicated engine, validated against an analytic or numerical reference, and scored by relative L_2 error. Across the seventeen cases with an archived reference-validated error, the result is a clean stratification. PINNs reach sub-percent accuracy on smooth, well-posed forward PDEs (a one-dimensional wave at 5×10^{-5} , a reaction-diffusion Allen-Cahn field at 1.2×10^{-3} , several mining and transport reduced models below 10^{-3}), with ten of seventeen cases under 1% relative error. But they degrade by two to three orders of magnitude on the hard classes: inverse problems (4×10^{-2}), high-wavenumber Helmholtz (1.0×10^{-1}), the lid-driven cavity Navier-Stokes benchmark (1.7×10^{-1}), operator learning (5.6×10^{-2}), and real noisy field data (6.9×10^{-2}). The finding matches the field’s own failure-mode literature: PINNs are an accurate and flexible instrument where the solution is smooth and the problem well-posed, and they are not a drop-in replacement for a good classical solver where it is not. Every case ships its equations, its method, a synthetic provenance label, and a reproducible error; every number and figure regenerates from the archived artifacts.

Index Terms. Physics-informed neural networks, scientific machine learning, partial differential equations, operator learning, inverse problems, benchmarking, relative L_2 error, reproducibility.

I. INTRODUCTION

PHYSICS-informed neural networks [1] solve a differential equation by training a network to satisfy its residual and boundary conditions at collocation points, and they have become a large and active corner of scientific machine learning [2]. The appeal is generality: the same recipe handles

forward and inverse problems, irregular domains, and sparse data. The catch, documented carefully by the field’s own analyses, is that a vanilla PINN often fails to train or converges to a wrong solution on precisely the problems that matter (stiff systems, sharp gradients, high frequencies) [3], [4], which is why a ladder of methods has grown to fix each failure mode: hard-constraint architectures, residual-based adaptive sampling [5], Fourier-feature and sinusoidal encodings for high frequencies, domain decomposition [6], and operator learning that amortises the solve over a family of inputs [7], [8]. What is missing from most presentations is a single answer to the practical question: on a given kind of problem, with the right method from that ladder, how accurate is a PINN in practice?

This report answers it with a runnable catalogue, PINN-Lab. Each case is a differential-equation problem paired with the state-of-the-art method that suits it; the case is trained offline by a dedicated engine (built on the standard tooling [9]), validated against an analytic solution (by the method of manufactured solutions) or a trusted numerical reference, exported to a portable format, and scored by relative L_2 error against that reference. The catalogue spans the method ladder (hard constraints and adaptive sampling on canonical PDEs, operator learning on a Darcy flow, Hamiltonian learning on a pendulum, conformal and Bayesian uncertainty on transport, and inverse recovery of hidden fields), so the atlas is not one method’s best case but a cross-section of where the whole ladder stands. The contribution is not a new method; it is a reproducible map of achieved accuracy, and the stratification it reveals.

II. THE CATALOGUE AND ITS PROTOCOL

A case is a differential equation, a method, a reference, and a metric. The equation ranges over canonical PDEs (a one-dimensional wave and heat equation, Burgers, Allen-Cahn, Poisson, the lid-driven cavity Navier-Stokes benchmark [10]), inverse problems (recovering a diffusivity field or a hidden velocity from sparse observations), dynamical systems (a Hamiltonian pendulum), and applied reduced models drawn from mining, pollution transport and environmental physics. The method is drawn from the state-of-the-art ladder and chosen to suit the case: adaptive residual sampling [5] for

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Runnable catalogue and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_PINNLAB; the catalogue is available at <https://pinnlab.fasl-work.com>.

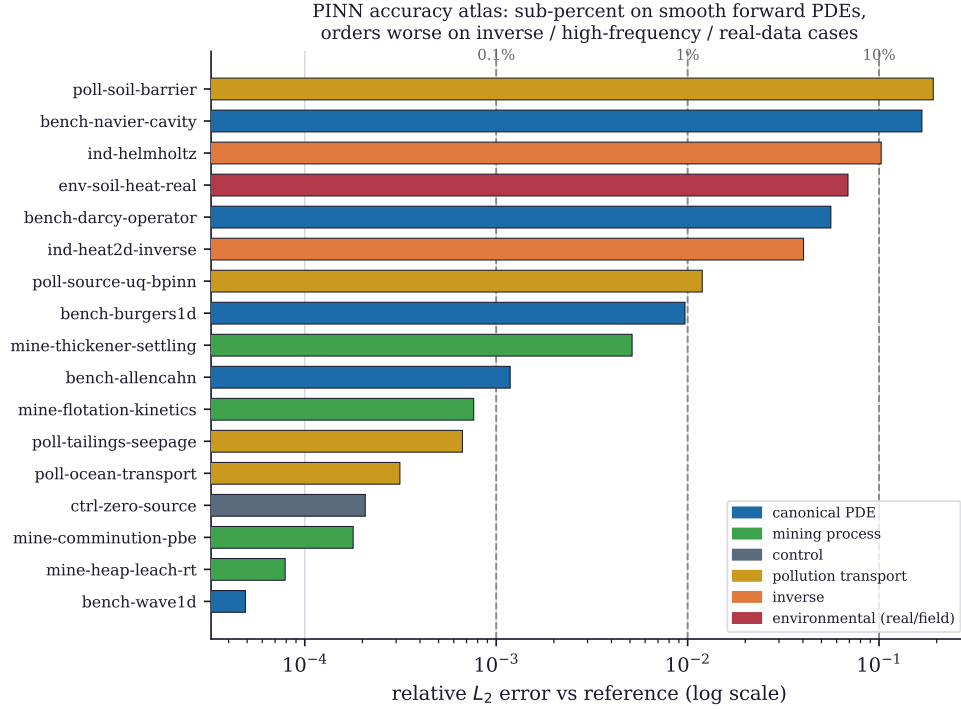


Fig. 1. The relative L_2 error of every reference-validated case, on a log axis, coloured by problem category, with reference lines at 0.1%, 1% and 10%. The stratification is clean: smooth well-posed forward PDEs and applied reduced models sit at the bottom (sub-percent, the best at 5×10^{-5}), while inverse problems, high-wavenumber Helmholtz, the lid-driven cavity, operator learning and real field data cluster at the top (4×10^{-2} to 1.9×10^{-1}). Where a PINN is accurate and where it is not is not random; it tracks the smoothness and well-posedness of the problem.

sharp fronts, a Fourier neural operator [7] for the parametric Darcy flow, a Hamiltonian network for the conservative pendulum, a Bayesian PINN for the uncertainty-quantified transport source. Each case is trained under a fixed seed, so a run is deterministic in (*case*, *seed*). The reference is an analytic solution where one exists (the method of manufactured solutions gives an exact target and an exact error), and a trusted numerical solution otherwise; the metric is the relative L_2 error of the PINN field against that reference, reported per case, with each case carrying a synthetic or synthetic-illustrative provenance label so no reduced model is mistaken for a calibrated digital twin. The reference-validated error is the number that matters, and it is archived alongside the field.

III. THE ACCURACY ATLAS

The atlas is Fig. 1 and Table I, and its shape is the report’s main result. Ten of the seventeen reference-validated cases reach relative L_2 error below 1%, and fourteen below 10%; the median is 5×10^{-3} . The accurate end holds what PINN theory predicts: smooth, well-posed forward problems with analytic references, the one-dimensional wave at 4.9×10^{-5} , several mining and transport reduced models (heap-leach reactive transport 7.9×10^{-5} , comminution population balance 1.8×10^{-4} , ocean transport 3.1×10^{-4}), and the nonlinear Allen-Cahn field at 1.2×10^{-3} . The inaccurate end holds the known-hard classes: the lid-driven cavity Navier-Stokes benchmark at 1.7×10^{-1} , high-wavenumber Helmholtz at 1.0×10^{-1} , the soil-barrier transport with a sharp front at

TABLE I
THE REFERENCE-VALIDATED CASES, SORTED BY RELATIVE L_2 ERROR. SMOOTH FORWARD PDES AND APPLIED REDUCED MODELS REACH SUB-PERCENT ERROR; INVERSE, HIGH-FREQUENCY, OPERATOR AND REAL-DATA CASES ARE ORDERS WORSE.

Case	category	rel. L_2
bench-wave1d	canonical PDE	4.9×10^{-5}
mine-heap-leach-rt	mining process	7.9×10^{-5}
mine-comminution-pbe	mining process	1.8×10^{-4}
ctrl-zero-source	control	2.1×10^{-4}
poll-ocean-transport	pollution transport	3.1×10^{-4}
poll-tailings-seepage	pollution transport	6.7×10^{-4}
mine-flotation-kinetics	mining process	7.6×10^{-4}
bench-allencahn	canonical PDE	1.2×10^{-3}
mine-thickener-settling	mining process	5.1×10^{-3}
bench-burgers1d	canonical PDE	9.7×10^{-3}
poll-source-uq-bpinn	pollution (UQ)	1.2×10^{-2}
ind-heat2d-inverse	inverse	4.0×10^{-2}
bench-darcy-operator	operator (FNO)	5.6×10^{-2}
env-soil-heat-real	real field data	6.9×10^{-2}
ind-helmholtz	high-frequency	1.0×10^{-1}
bench-navier-cavity	Navier-Stokes	1.7×10^{-1}
poll-soil-barrier	sharp-front transport	1.9×10^{-1}

1.9×10^{-1} , an inverse heat-conduction problem at 4.0×10^{-2} , the Fourier-neural-operator Darcy solve at 5.6×10^{-2} , and the one case fit to *real* soil temperature field data at 6.9×10^{-2} . The gap of two to three orders of magnitude between the ends quantifies, across one consistent catalogue, the documented failure-mode picture of PINNs [3], [4]: accurate on smooth forward problems and, on stiff, high-frequency, inverse or noisy problems, at an accuracy a classical solver would

Allen-Cahn (nonlinear reaction-diffusion), relative $L_2 \approx 1.2\text{e-}3$

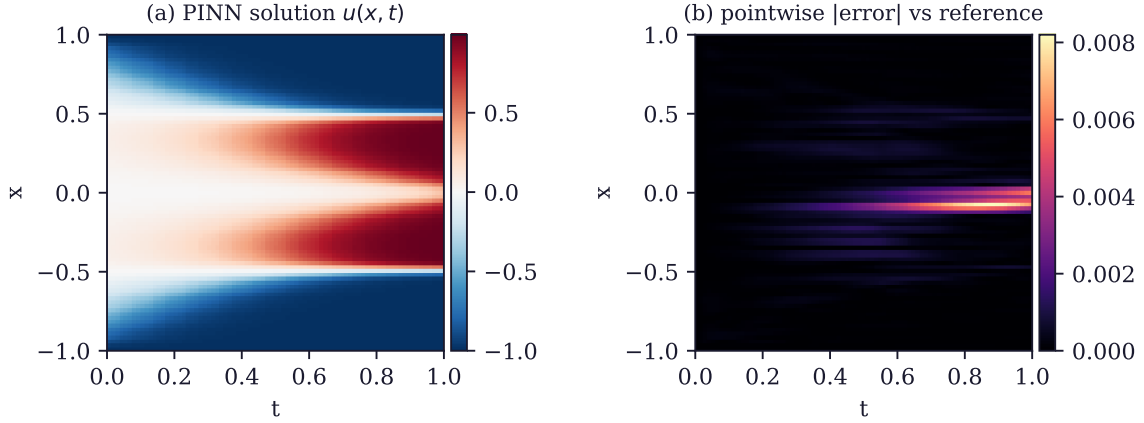


Fig. 2. A well-solved case, the nonlinear Allen-Cahn reaction-diffusion equation (relative $L_2 \approx 1.2 \times 10^{-3}$), trained with residual-based adaptive sampling. (a) The PINN solution $u(x, t)$ reproduces the phase-separation dynamics. (b) The pointwise error against the reference is near zero almost everywhere and concentrates at the sharp interface near $x = 0$ and late time, the steep-gradient region where a PINN’s smoothness prior is most strained, exactly where the ladder’s adaptive-sampling and hard-constraint methods are designed to help.

consider unacceptable.

IV. A SOLVED CASE, AND WHERE THE ERROR LIVES

Figure 2 shows what a good case looks like and where even a good case errs. Allen-Cahn is a stiff nonlinear reaction-diffusion equation whose solution separates into phases with a thin moving interface; the PINN, trained with residual-based adaptive sampling that places extra collocation points where the residual is largest [5], reaches 1.2×10^{-3} relative L_2 . The error field is the informative part: it is essentially zero across the smooth bulk and concentrates at the sharp interface at late time, the steep-gradient region where a smooth network struggles to represent a near-discontinuity. This is the microcosm of the whole atlas. A PINN is accurate wherever the solution is smooth, and its error localises exactly at the sharp, high-frequency or ill-posed features, which is why the cases dominated by such features (Helmholtz, the cavity, sharp fronts, inverse recovery) sit at the inaccurate end of Fig. 1. The method ladder exists to push that localised error down, and it does: in the comparison run stored with this case, the relative L_2 error against the spectral reference solution is 0.954 for a naive soft-constraint PINN and 4.34×10^{-3} for a separately trained hard-constraint PINN with residual-based adaptive sampling, a factor of 219.8, more than two orders of magnitude. The ladder does not, however, erase the fundamental difficulty.

V. DISCUSSION

The atlas bounds what a PINN delivers in both directions. The two-to-three-order accuracy gap tracks problem smoothness and well-posedness, and on the classic hard benchmarks (lid-driven cavity, high-wavenumber Helmholtz) the achieved error is far from what a classical finite-element or finite-volume solver delivers, so a PINN is not a replacement for such a solver on a single well-posed forward solve. At the

same time, ten of the seventeen cases reach sub-percent error. The problems where classical solvers are awkward (inverse problems where the unknown is a field, not a boundary condition; assimilation of sparse or real data into a physical model; operator learning that amortises a family of solves) are also the harder cases in the atlas. The trade-off is therefore between forward accuracy and the ability to pose a problem that a classical solver cannot pose cleanly, and the atlas shows, per problem type, which side of it applies, with the measured error.

VI. SCOPE AND LIMITATIONS

This is a catalogue of method *types* exercised at lab scale, not an exhaustively tuned leaderboard; a specialist could push any single case lower with more training and architecture search, and the atlas reports the as-catalogued accuracy, not a claimed optimum. Most cases are validated on analytic (method-of-manufactured-solutions) anchors or faithful reduced models, each labelled *synthetic* or *synthetic-illustrative*; the reduced mining and pollution models are physically motivated but are not calibrated digital twins of any operation, and the one real-data case is labelled as such. A good classical solver beats a PINN on a single well-posed forward problem, and PINN-Lab says so; it is a teaching and method instrument, not a production solver. The dynamical and operator cases use error metrics suited to their outputs (trajectory and sample errors), reported in their own artifacts. And, as an exploratory laboratory, its aim is to map where each method type applies, not to claim a method novel beyond the state of the art. The catalogue states the same scope on every panel.

VII. CONCLUSION

Exercised across one consistent, reference-validated catalogue, physics-informed neural networks show a clean strat-

ification: sub-percent relative L_2 error on smooth, well-posed forward PDEs and applied reduced models (the best at 5×10^{-5} , ten of seventeen cases under 1%), and two-to-three orders worse on the hard classes: inverse problems, high-wavenumber and sharp-gradient PDEs, operator learning and real noisy data. The error localises exactly at the sharp and ill-posed features, matching the field’s failure-mode theory. The resulting map (PINNs are an accurate and flexible instrument where the solution is smooth and the problem well-posed, and not a replacement for a good classical solver where it is not) is the report’s contribution, made reproducible with every case’s equations, method, provenance label and error archived for inspection.

APPENDIX A REPRODUCTION

Every case’s reference-validated error and fields are archived under `data/derived/`, one folder per case: a `field.json` holds the solution and its `l2_relative` summary, and for 14 of the 17 cases a `comparison.json` holds the method variants and their error fields. The atlas summary `manuscripts/pinn-atlas/data/atlas.json` collects the `l2_relative` value of every case, and the figure script `manuscripts/pinn-atlas/figures/make_figs.py` draws the two figures from that summary and from the Allen-Cahn case artifacts. Because each case is deterministic in `(case, seed)`, re-running reproduces the errors; Table I and Figs. 1–2 regenerate from the archived artifacts with no retraining.

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